Can a Connection Remember?

Synaptic Plasticity as Short-Term Memory · DataForge 2026 · Pathway Track — Explain the Frontier

Central claim

Short-term synaptic plasticity can store recent information in changing synaptic states, but trace persistence creates a trade-off between retaining older information and responding to newer information. Our explainer turns this into a small, testable experiment: a learner teaches an association, waits while its trace decays, introduces another association, predicts which trace will dominate at readout, and then changes persistence and delay to test the prediction.

Mechanism

In our educational model, a synaptic connection is represented by a scalar state *w*. Learning updates it as **$w(t+1) = \delta w(t) + \eta \cdot pre \cdot post$**, where η controls new-write strength and δ controls persistence. During a delay, an old trace therefore decays approximately as $\delta^{\Delta t}$. Recall selects the strongest current association. This is a transparent abstraction, not a biological law: we do not simulate membrane voltage, neurotransmitter dynamics, calcium, STDP, or detailed spiking. A synapse is not claimed to literally store a human memory. Apple→Red and Apple→Green are separate traces; Green does not erase Red. Competition occurs when their current strengths are compared at readout.

Why this mechanism matters

Conventional Transformers normally keep learned parameters fixed during inference and represent transient context through activations/attention state. Dynamic synaptic-state approaches add an explicitly evolving internal state that can be written, persist, decay, and influence later computation. The benefit is a compact way to study recency and adaptation; the trade-off is that stronger persistence can preserve stale information, while stronger plasticity can make the system more responsive but less stable. The toy exposes this trade-off directly rather than hiding it behind a large model.

Evidence and research landscape

STPN (Garcia Rodriguez et al., 2022) gives a machine-learning example in which synapses carry state through time; the reported STPN outperformed tested RNN, LSTM, fast-weight and differentiable-plasticity alternatives across several supervised and reinforcement-learning tasks. [1] **STSP working memory (Kozachkov et al., 2022)** compared recurrent networks with and without STSP: both maintained memories despite distractors, while STSP networks showed more brain-like activity and greater robustness to network degradation in their experiments. [2] **Chrysanthidis et al. (2025)** used a richer spiking model with multiple synaptic timescales to study how short-term traces interact with episodic-memory recall and recency. [3] These are computational-model results, not proof that one mechanism universally explains biological working memory. Our Apple experiment is an educational simplification, not a reproduction of any paper.

BDH and BDH-CQ

BDH / Dragon Hatchling (Kosowski et al., 2025) is the direct frontier-AI connection. Its authors describe working memory during inference as relying on synaptic plasticity with Hebbian learning in a brain-inspired, graph-based language-model architecture, with individual synapses strengthening around concepts encountered during processing. [4] Our toy isolates the intuition that changing synaptic state can carry recent information; it does **not** implement BDH. **BDH-CQ (Engdahl et al., 2026)** extends the family toward in-context learning and recurrent latent reasoning: inference-time examples update recurrent memory, followed by iterative latent computation without a verbalized chain-of-thought. [5] A 150M-parameter BDH-CQ configuration reported 29.5% pass@2 on ARC-AGI-1 at a computed $0.00070/task; Pathway also reports that an independent black-box evaluation of the deployed system reproduced the result. [6] This is evidence about BDH-CQ's benchmark/deployment evaluation, not evidence that our scalar rule is sufficient for reasoning.

Contribution, maturity, and limits

The submission is a live Streamlit explainer backed by a deterministic Python model. The model performs learning, decay, recall, and the final scenario computation; the interface exposes numerical state, a live synaptic visualization, trajectories, prediction prompts, and the final computed outcome. No pretrained weights or external dataset determine the learner's result; the examples are synthetic and the visualization is driven by computed state. The claim is reproducible in under a minute. The main limitation is deliberate compression: scalar weights, a simple argmax readout, and a simplified Hebbian update omit the high-dimensional dynamics and biophysics of research systems. Short-term plasticity is a mature research concept, but its use as a general-purpose memory substrate for frontier language models remains an emerging question. The most important open issue is whether richer synaptic-state mechanisms can scale while preserving accuracy, stability, interpretability, and efficiency. Our artifact demonstrates the mechanism's computational intuition, not a biological or frontier-model proof.

Primary sources / further reading

[1] Garcia Rodriguez, H., Guo, Q., & Moraitis, T. (2022). *Short-Term Plasticity Neurons Learning to Learn and Forget*. ICML/PMLR 162. https://proceedings.mlr.press/v162/rodriguez22b.html

[2] Kozachkov, L. et al. (2022). *Robust and brain-like working memory through short-term synaptic plasticity*. PLOS Computational Biology 18(12), e1010776. https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010776

[3] Chrysanthidis, N. et al. (2025). *Short-term plasticity influences episodic memory recall*. Scientific Reports 15, 28164. https://www.nature.com/articles/s41598-025-12611-5

[4] Kosowski, A. et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507. https://arxiv.org/abs/2509.26507

[5] Engdahl, B. et al. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888. https://arxiv.org/abs/2608.09888

[6] Pathway (2026). *Reasoning at a fraction of the compute* — reports the deployed-system evaluation and ARC-AGI-1 result. https://pathway.com/research/introducing-bdh-cq